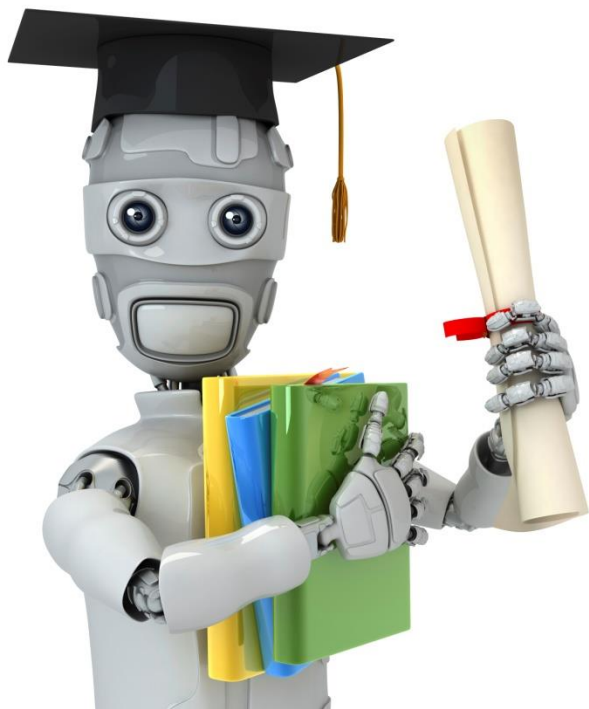


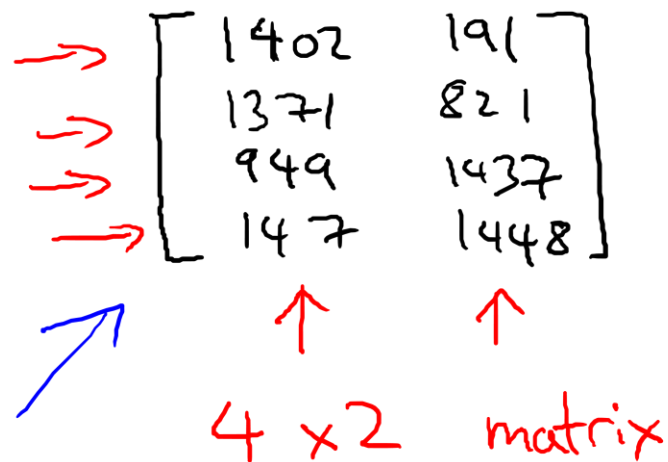


Machine Learning

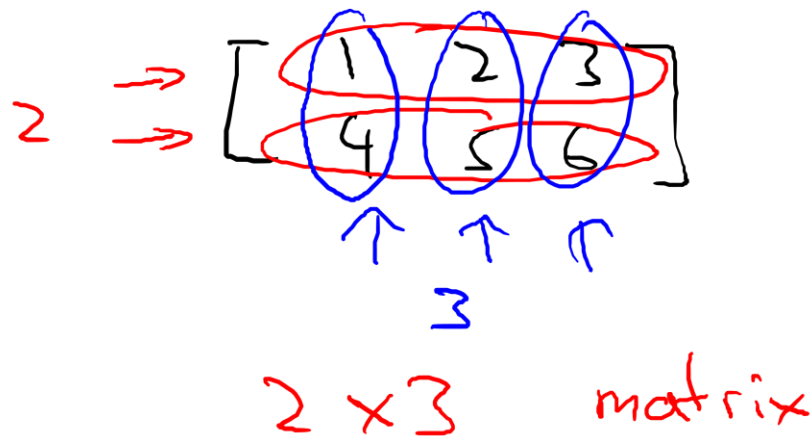
Linear Algebra
review (optional)

Matrices and
vectors

Matrix: Rectangular array of numbers:



→ $\mathbb{R}^{4 \times 2}$



$\mathbb{R}^{2 \times 3}$

Dimension of matrix: number of rows x number of columns

Matrix Elements (entries of matrix)

$$A = \begin{bmatrix} 1402 & 191 \\ 1371 & 821 \\ 949 & 1437 \\ 147 & 1448 \end{bmatrix}$$

A_{ij} = " i, j entry" in the i^{th} row, j^{th} column.

$$A_{11} = 1402$$

$$A_{12} = 191$$

$$A_{32} = 1437$$

$$A_{41} = 147$$

$$\cancel{A_{43}} = \text{Undefined (error)}$$

Vector: An $n \times 1$ matrix.

$$\underline{y} = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$$

$\uparrow \quad \uparrow$
 $n = 4$

$\leftarrow$ 4-dimensional vector.

~~$\mathbb{R}^{3 \times 2}$~~

$\mathbb{R}^4$

$y_i = i^{th}$ element

$$y_1 = 460$$

$$y_2 = 232$$

$$y_3 = 315$$

$\rightarrow$ A, B, C, X

a, b, x, y

1-indexed vs 0-indexed:

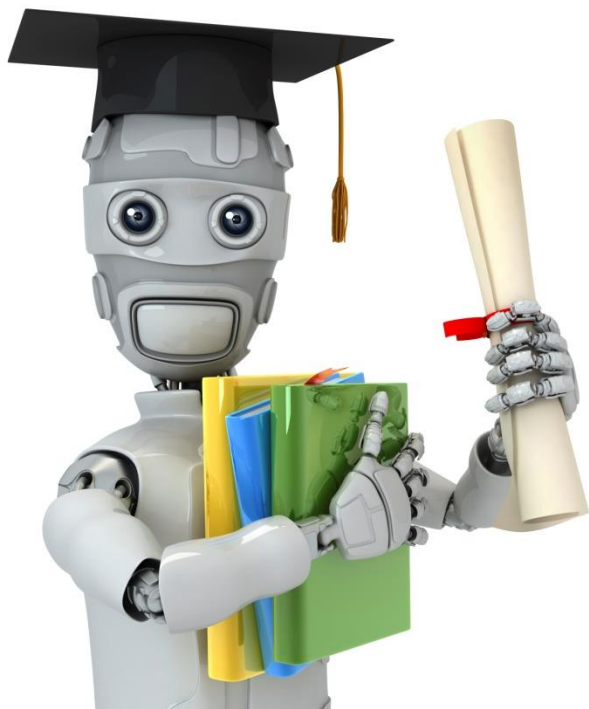
$y[1]$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

$\uparrow$ 1-indexed

$$y = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

0-indexed



Machine Learning

Linear Algebra review (optional)

Addition and scalar multiplication

Matrix Addition

$$\begin{array}{c} \downarrow \quad \downarrow \\ \rightarrow \left[\begin{array}{cc} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{array} \right] + \left[\begin{array}{cc} 4 & 0.5 \\ 2 & 5 \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} 5 & 0.5 \\ 4 & 10 \\ 3 & 2 \end{array} \right] \\ \text{3} \times \text{2} \quad \text{3} \times \text{2} \quad \text{3} \times \text{2} \\ \text{matrix} \end{array}$$

$$\begin{array}{c} \rightarrow \left[\begin{array}{cc} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{array} \right] + \left[\begin{array}{cc} 4 & 0.5 \\ 2 & 5 \end{array} \right] = \\ \text{3} \times \text{2} \quad \text{2} \times \text{2} \quad \text{error} \end{array}$$

Scalar Multiplication

← real number

$$3 \times \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 6 & 15 \\ 9 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} \times 3$$

3x2 3x2

$$\begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} / 4 =$$

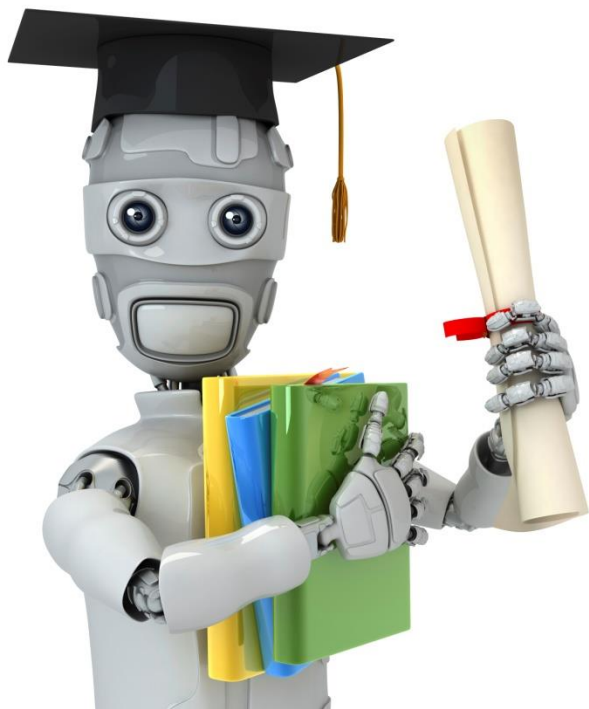
$$\frac{1}{4} \begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 \\ \frac{3}{2} & \frac{3}{4} \end{bmatrix}$$

Combination of Operands

$$\begin{aligned}
 & \text{Scalar multiplication} \rightarrow 3 \times \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} \text{ /3} \quad \text{Scalar division} \\
 & = \begin{bmatrix} 3 \\ 12 \\ 6 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ \frac{2}{3} \end{bmatrix} \quad \begin{array}{l} \text{matrix subtraction /} \\ \text{vector subtraction} \end{array} \\
 & = \begin{bmatrix} 2 \\ 12 \\ 10\frac{1}{3} \end{bmatrix} \quad \begin{array}{l} \text{matrix addition /} \\ \text{vector addition} \end{array}
 \end{aligned}$$

3×1 matrix
 3 -dimensional vector



Machine Learning

Linear Algebra review (optional)

Matrix-vector multiplication

Example

$$\begin{bmatrix} 1 & 3 \\ 4 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 16 \\ 4 \\ 7 \end{bmatrix}$$

3×2 2×1 3×1 matrix

$$1 \times 1 + 3 \times 5 = 16$$

$$4 \times 1 + 0 \times 5 = 4$$

$$2 \times 1 + 1 \times 5 = 7$$

Details:

$$\begin{array}{ccc} \underline{A} & \times & \underline{x} \\ \begin{array}{c} \text{[Diagram of matrix A with m rows and n columns, highlighted in pink, green, blue, and black]} \\ \text{[Diagram of vector x with n elements, highlighted in pink, green, blue, and black]} \end{array} & = & \underline{y} \\ \begin{array}{c} \text{[Diagram of vector y with m elements, highlighted in pink, green, blue, and black]} \end{array} & & \end{array}$$

$\boxed{m} \times \boxed{n}$ matrix
(m rows,
 n columns)

$\boxed{n} \times 1$ matrix
(n -dimensional
vector)

$\boxed{m}$ -dimensional
vector

→ To get y_i , multiply A 's i^{th} row with elements of vector x , and add them up.

Example

$$\begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 3 & 0 & 4 \\ -1 & -2 & 0 & 0 \end{bmatrix}_{3 \times 4} \begin{matrix} \downarrow \\ 1 \\ 3 \\ 2 \\ 1 \end{matrix}_{4 \times 1} = \begin{bmatrix} 14 \\ 13 \\ -7 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 14 \\ 13 \\ -7 \end{bmatrix}$$

$$1 \times 1 + 2 \times 3 + 1 \times 2 + 5 \times 1 = 14$$

$$0 \times 1 + 3 \times 3 + 0 \times 2 + 4 \times 1 = 13$$

$$-1 \times 1 + (-2) \times 3 + 0 \times 2 + 0 \times 1 = -7$$

House sizes:

- 2104
- 1416
- 1534
- 852

Matrix

$$\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix}$$

4x2

$$h_{\theta}(x) = -40 + 0.25x$$

$h_{\theta}(x)$

2x1

Vector

$$\begin{bmatrix} -40 \\ 0.25 \end{bmatrix}$$

x

4x1 matrix

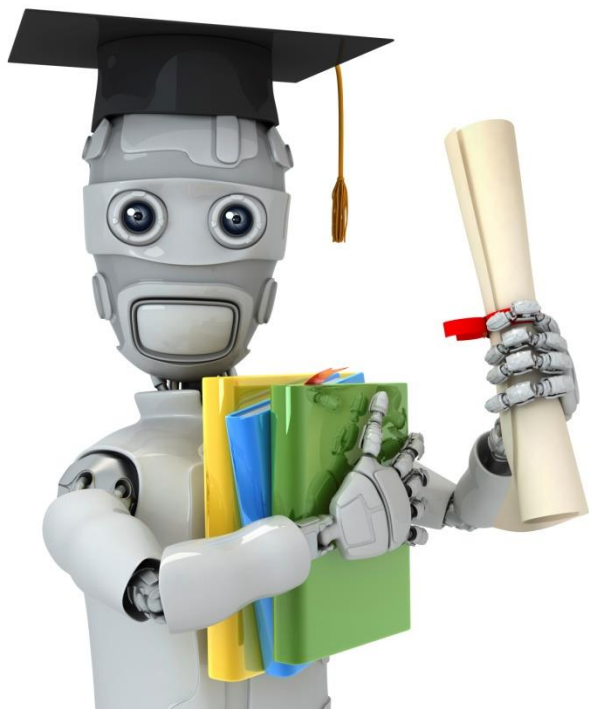
$$\begin{bmatrix} -40 \times 1 + 0.25 \times 2104 \\ -40 \times 1 + 0.25 \times 1416 \\ \\ \end{bmatrix}$$

$h_{\theta}(1416)$

$$\text{prediction} = \text{Data Matrix} \times \text{Parameters}$$

4x1

for $i = 1, \dots, 1000$,
prediction(i) = ...



Machine Learning

Linear Algebra review (optional)

Matrix-matrix multiplication

Example

$$\begin{array}{l} \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 0 & 1 \\ \hline 5 & 2 \\ \hline \end{array} = \begin{bmatrix} 11 & 10 \\ 9 & 14 \end{bmatrix} \\ \text{2} \times 3 \quad \text{3} \times 2 \\ \hline \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{array}{|c|} \hline 1 \\ \hline 0 \\ \hline 5 \\ \hline \end{array} = \begin{bmatrix} 11 \\ 9 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{array}{|c|} \hline 3 \\ \hline 1 \\ \hline 2 \\ \hline \end{array} = \begin{bmatrix} 10 \\ 14 \end{bmatrix} \end{array}$$

Diagram illustrating matrix multiplication with dimensions and intermediate results:

- Top row: $\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 0 & 1 \\ \hline 5 & 2 \\ \hline \end{array} = \begin{bmatrix} 11 & 10 \\ 9 & 14 \end{bmatrix}$. Dimensions 2×3 and 3×2 are indicated. The result matrix has elements 11, 10, 9, 14.
- Middle row: $\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{array}{|c|} \hline 1 \\ \hline 0 \\ \hline 5 \\ \hline \end{array} = \begin{bmatrix} 11 \\ 9 \end{bmatrix}$. The dimension 3×2 is indicated. The result matrix has elements 11, 9.
- Bottom row: $\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{array}{|c|} \hline 3 \\ \hline 1 \\ \hline 2 \\ \hline \end{array} = \begin{bmatrix} 10 \\ 14 \end{bmatrix}$. The dimension 3×2 is indicated. The result matrix has elements 10, 14.

Arrows indicate the mapping of dimensions and the resulting elements in the final matrix:

- Green arrows point from the 2×3 dimension to the first column of the result matrix (11, 9).
- Blue arrows point from the 3×2 dimension to the second column of the result matrix (10, 14).

Details:

$$\begin{array}{c}
 \underline{A} \quad \times \quad \underline{B} \quad = \quad \underline{C} \\
 \left[\begin{array}{c} \end{array} \right] \times \left[\begin{array}{c|c|c|c} & & & \end{array} \right] = \left[\begin{array}{c|c|c|c} & & & \end{array} \right]
 \end{array}$$

$m \times n$ matrix
 (m rows, n columns)

$n \times o$ matrix
 (n rows, o columns)

$m \times o$ matrix

~~$n \times 1$~~

The i^{th} column of the matrix C is obtained by multiplying A with the i^{th} column of B . (for $i = 1, 2, \dots, o$)

Example

$$\begin{matrix} 2 \times 2 & 2 \times 2 \\ \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} & \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} \end{matrix}$$

$$= \begin{bmatrix} 9 & 7 \\ 15 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 0 + 3 \times 3 \\ 2 \times 0 + 5 \times 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 1 + 3 \times 2 \\ 2 \times 1 + 5 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

House sizes:

$$\begin{cases} 2104 \\ 1416 \\ 1534 \\ 852 \end{cases}$$

Have 3 competing hypotheses:

1. $h_{\theta}(x) = -40 + 0.25x$

2. $h_{\theta}(x) = 200 + 0.1x$

3. $h_{\theta}(x) = -150 + 0.4x$

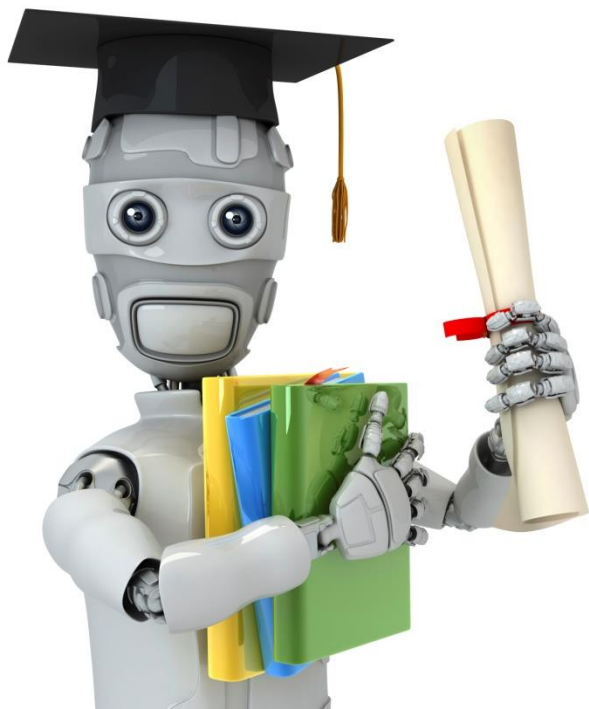
Matrix

Matrix

$$\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix} \times \begin{bmatrix} -40 & 200 & -150 \\ 0.25 & 0.1 & 0.4 \end{bmatrix} = \begin{bmatrix} 486 & 410 & 692 \\ 314 & 342 & 416 \\ 344 & 353 & 464 \\ 173 & 285 & 191 \end{bmatrix}$$

Prediction
of first
 h_{θ}

Predictions
of 2nd
 h_{θ}



Machine Learning

Linear Algebra review (optional)

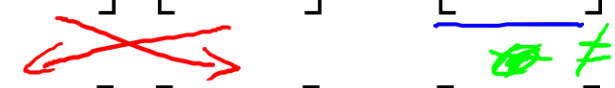
Matrix multiplication properties

$$3 \times 5 = 5 \times 3$$


"Commutative"

Let A and B be matrices. Then in general,
 $A \times B \neq B \times A$. (not commutative.)

E.g.

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \quad \left| \quad \begin{array}{l} A \times B \\ m \times n \quad n \times m \end{array} \right.$$


$$\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \end{bmatrix} \quad \left| \quad \begin{array}{l} B \times A \\ n \times n \end{array} \right.$$


$A \times B$ is $m \times m$
 $B \times A$ is $n \times n$

$$\underline{3 \times 5 \times 2}$$

$$3 \times 10 = 30 = 15 \times 2$$

$$3 \times (5 \times 2) = (3 \times 5) \times 2$$

"Associative"

$$\begin{array}{l} A \times (B \times C) \quad \leftarrow \\ \underline{(A \times B)} \times C \quad \leftarrow \end{array}$$

$$A \times B \times C.$$

Let $D = B \times C$. Compute $A \times D$.

Let $E = A \times B$. Compute $E \times C$.

$$A \times (B \times C)$$

$$(A \times B) \times C$$

Some
answer.

Identity Matrix

1 is identity.

$$1 \times z = z \times 1 = z$$

for any z

Denoted I (or $I_{n \times n}$).

Examples of identity matrices:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}_{1 \times 1}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4}$$

Informally:

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

For any matrix A ,

$$A \cdot I = I \cdot A = A$$

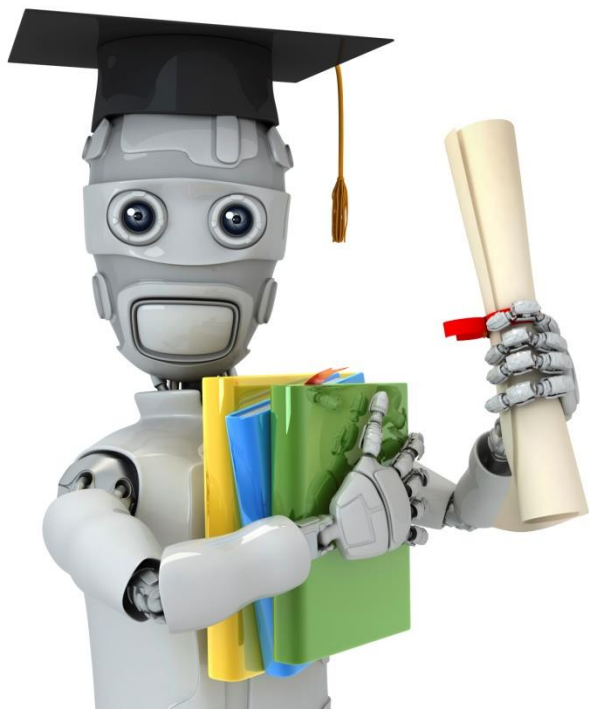
$m \times n$ $n \times n$ $m \times m$ $m \times n$ $m \times n$

$$I_{n \times n}$$

Note:

$$AB \neq BA \text{ in general}$$

$$AI = IA \checkmark$$



Machine Learning

Linear Algebra
review (optional)

Inverse and
transpose

I = "identity."

$$3 \underbrace{(3^{-1})}_{\frac{1}{3}} = 1$$

$$12 \times \underbrace{(12^{-1})}_{\frac{1}{12}} = 1$$

$$0 \underbrace{(0^{-1})}_{\text{undefined}}$$

Not all numbers have an inverse.

Matrix inverse: ↖ square matrix
(#rows = #columns) A^{-1}

If A is an $m \times m$ matrix, and if it has an inverse,

$$\rightarrow \underline{A(A^{-1})} = \underline{A^{-1}A} = \underline{I}.$$

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

↑

e.g.

$$\underbrace{\begin{bmatrix} 3 & 4 \\ 2 & 16 \end{bmatrix}}_{A \text{ (2x2)}} \underbrace{\begin{bmatrix} 0.4 & -0.1 \\ -0.05 & 0.075 \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{A^{-1}A} = I_{2 \times 2}$$

Matrices that don't have an inverse are "singular" or "degenerate"

Matrix Transpose

Example:

$$\underline{A} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 5 & 9 \end{bmatrix}$$

Handwritten annotations: A red arrow points to the underlined A. A blue arrow points from the first row to the first column. A green oval encloses the first row [1, 2, 0]. A blue oval encloses the second row [3, 5, 9]. A red '2 x 3' is written below the matrix.

$$\underline{B} = \underline{A}^T = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 9 \end{bmatrix}$$

Handwritten annotations: A blue arrow points to the underlined B. A green oval encloses the first column [1, 2, 0]. A blue oval encloses the second column [3, 5, 9]. A red '3 x 2' is written below the matrix.

Let A be an $m \times n$ matrix, and let $B = A^T$.

Then B is an $n \times m$ matrix, and

$$\underline{B}_{ij} = \underline{A}_{ji}.$$

Handwritten annotations: Red underlines are under the i and j in both B and A.

$$B_{12} = A_{21} = 2$$

$$B_{32} = 9$$

$$A_{23} = 9.$$